

Education ERP

User Guide · Admin Guide · Demo Script

Audience: guardians, applicants, students, school staff, and demo presenters

Platform: Frappe Desk (staff) + bilingual Vue portals (families)

Custom app: university_erp on Frappe v16, ERPNext v16, Frappe Education, and Frappe CRM

Status: staging-ready foundations. Human UAT and live providers are still deferred.

Important: The product is not production-ready. Payments and scans use fake adapters. Money is not taken from a real bank account. OTP test code is 246810.

This file is the single operating guide. It describes what is implemented today, who does each step, the exact record types involved, and a timed demo flow you can present without inventing screens that do not exist.

Prepared 2026-08-17 · Matches implemented university_erp portals and DocTypes through Phase 8 staging.

1. How to use this document

You are...	Start at
A guardian or student	Part A — User Guide
School staff (admissions, academics, finance, registrar)	Part B — Admin Guide
Presenting the product	Part C — Demo Presentation Flow
Looking up a screen or DocType	Part D — Feature catalogue

Staff work in Frappe Desk. Families work in two public pages:

Surface	Path	Who
Guardian admission portal	/guardian-admission	Parent/guardian applying for a child
Student / guardian portal	/student-portal?access=<token>	Enrolled student or guardian with a time-limited link
Staff Desk	/login then Desk search	Trained staff

Staging example (Railway): <https://web-production-7580e.up.railway.app>. Local development typically uses [erp.localhost](#) or [p21.localhost](#).

Do not put passwords in this guide. Desk credentials live in the operator secret store ([SITE_ADMIN_PASSWORD](#) / local [.env](#)). Never demonstrate production credentials on a projector.

2. What the product does

The Education ERP runs one connected admission-to-fee journey for an Indian school or college:

1. The institution configures its hierarchy, academic year, program, class, intake, fee policy, and published application form.
2. A guardian starts an application on a phone, in English or Hindi.
3. The system creates a CRM enquiry and a saveable draft.
4. The guardian uploads documents and starts the application fee.
5. Staff hand the enquiry to a Student Applicant, check eligibility, publish merit, and allocate seats without overselling capacity.
6. After documents and fees pass, staff confirm admission and convert the applicant into exactly one Student and exactly one Program Enrollment.
7. Finance generates a fee demand that posts a Sales Invoice in ERPNext, collects online or offline payment, issues a receipt, and can refund and reconcile to the General Ledger.
8. The family later opens a scoped student portal to see dues, receipts, documents, and notices.

ERPNext remains the accounting source of truth. Custom records hold education rules; they do not keep a second ledger.

2.1 Who uses it

Persona	Typical tool	Main jobs
Guardian / applicant	/guardian-admission	Register, apply, upload, pay application fee, see status
Student / guardian (enrolled)	/student-portal	View dues, pay fees, download receipts, read notices
Institution Administrator	Desk	Institution tree, users, published structure

Persona	Typical tool	Main jobs
Academic Officer	Desk	Year, program, class, section, timetable, intake
Admissions Officer / counsellor	Desk + CRM	Enquiry, handoff, documents, eligibility
Merit Operator / Approver	Desk	Merit configuration, publish run, seat offers
Registrar	Desk	Confirmation, conversion, identity, corrections
Finance Officer / Cashier	Desk	Demands, collection, receipts, refunds, reconciliation
Auditor / System Manager	Desk	Audit, exports, roles, technical health

2.2 What is in the current build vs later

In the current build (can be shown in a demo):

- Institution and academic masters, including timetable clash rejection
- Student identity, documents, consent, correction, duplicate review, privacy export request
- CRM lead application handoff Student Applicant
- Guardian PWA: bilingual steps, local autosave, server draft, document scan, application-fee order
- Eligibility, immutable merit, seat matrix, offers, waitlist, capacity lock
- Admission confirmation gates and idempotent student conversion
- Fee policy, demand + Sales Invoice, online/offline payment, receipt, refund, settlement import, GL check
- Student portal snapshot, receipt PDF, payment retry, fake OTP contract
- Role matrix, identifier masking, webhook replay rejection, correlation IDs

Deferred (do not claim these in a live demo):

- Real Razorpay / MSG91 / SMTP / Cloudflare R2 traffic
- Daily attendance, examinations, LMS, hostel, transport, library, HR
- Full Frappe Role Permission Manager sign-off and browser UAT
- Hindi human review and low-literacy usability sign-off
- Production DNS, signed image publish, production cutover

Payments and scans in staging use fake adapters. Money is not taken from a real bank account. OTP in the student portal is the local test code **246810**.

3. End-to-end product flow

This is the single story the product implements. Every later section is a zoom-in on one box.



Hard rules that must never be broken in a demo or in production:

- Accepted seats cannot exceed the seat-matrix capacity.
- One provider payment maps to at most one posted accounting result.
- Conversion creates at most one Student and one enrollment.
- A published merit run cannot be cancelled or overwritten.
- Fee demand net amount must equal the submitted Sales Invoice.

Part A — User Guide

This part is for families. Staff should still read it so they can help a guardian on a phone.

A1. Guardian admission portal

Open: </guardian-admission> Languages: English and हिन्दी (toggle in the header) Device: mobile-first. Works as a PWA (can be added to the home screen). Network: if the phone goes offline, the draft stays on the device. When the network returns, it syncs to the school.

A1.1 Before you start

Have ready:

- Guardian 10-digit mobile number
- Guardian name and child name
- Child date of birth and home address

- Previous school name (if any)
- Birth certificate (PDF, JPG, or PNG, under 5 MB)
- Child photo (PDF, JPG, or PNG, under 5 MB)
- A way to pay the ₹500 application fee (in staging this is a simulated payment)

The portal only appears if staff have published at least one Admission Application Form Version. If you see “No published admission form is available,” the school has not opened admissions yet.

A1.2 The six steps

The left (or top) step list is the whole journey. You can go back to a finished step. You cannot skip ahead.

Step 1 — Register (mobile)

1. Enter a 10-digit mobile number.
2. Enter the guardian name.
3. Enter the child’s name.
4. Tap Next.

The first successful save creates:

- a CRM Lead (enquiry) for the child
- an Admission Application Draft
- a resume token stored only on this phone (the school stores a hash, not the raw token)

If you close the browser and come back on the same phone, the form reloads from local storage and resumes the same draft.

Validation: all three fields are required; mobile must be exactly 10 digits.

Step 2 — Class

1. Choose the class applying for: Class 6, 7, 8, or 9 (pilot list).
2. Optionally enter the previous school.
3. Tap Next.

Step 3 — Details

1. Enter date of birth.
2. Enter home address.
3. Tick the consent box: the school may check these details for admission.
4. Tap Next.

Validation: date of birth, address, and consent are required.

Step 4 — Documents

1. Upload Birth certificate.
2. Upload Child photo.
3. Wait until each file is accepted.

What happens in the background:

- the file is stored as a private object
- a malware scan runs (fake ClamAV in staging)
- only Scan Passed files unlock the next step
- PDF, JPG, and PNG only; maximum 5 MB

If upload fails, stay on this step and try again. Do not create a second application.

Step 5 — Payment (₹500)

1. Read the safety message: pay once. If money is deducted, do not pay again.
2. Tap Pay ₹500. This starts a payment order and does not create a second order if you tap again.
3. If the button shows that payment has started, tap Check payment status instead of paying again.
4. Continue only when the portal shows payment recorded.

The amount is fixed at ₹500 by the published pilot policy. Other amounts are rejected by the server.

Step 6 — Status

You see a short summary:

- child name
- class
- documents complete / pending
- payment complete / pending

Message: the application is saved; the school will check details and update this page.

Tap Finish.

A1.3 What the guardian does not do

The guardian portal does not currently:

- log in with a password or OTP (access is the phone + resume token)
- accept or reject a seat offer
- see merit rank
- pay tuition (that is the student portal after enrollment)
- switch between multiple children on one account (one draft per device session)

Staff complete eligibility, merit, offer, confirmation, and conversion in Desk.

A1.4 If something goes wrong

What you see	What to do
Offline banner	Keep filling. The phone copy is saved. Sync retries when online.
“Phone copy saved; school sync will retry”	Stay on the same phone. Do not start a new application.
Document rejected / scan failed	Upload a clearer PDF/JPG/PNG under 5 MB.
Payment started, not complete	Use Check payment status. Do not pay again.
Form will not go Next	Read the red error: missing mobile, consent, documents, or payment.
New phone / cleared browser	Ask the school to help. The resume token lived on the old phone.

A2. Student and guardian portal (after admission)

Open: `/student-portal?access=<access-token>` The school issues a time-limited Student Portal Access link. Bookmarking the URL on the same phone stores the token locally.

If the token is missing, expired, or revoked, the page says the link is invalid and you must request a new one.

A2.1 What you see

Section	Meaning
Student	Child name and student ID. Access expiry date.
Fee dues	Open Student Fee Demand rows in status Generated , with amount and due date.
Receipts	Posted Student Fee Payment rows. Download saves a PDF.
Notices	Published Student Portal Notice rows for All Students or Guardians.
Documents	Student documents with verification status and expiry (if any).

Language toggle: English / हिन्दी.

A2.2 Pay a due

1. On a due row, tap Pay.
2. The portal creates one provider order for that demand. Tapping again reuses the same order.
3. After the (staging) capture, staff-side accounting posts one Payment Entry and one Student Fee Payment.
4. Refresh or reopen the portal to see the new receipt.

If the invoice is already fully paid, the server refuses a second charge.

A2.3 Download a receipt

Tap Download on a receipt. The PDF is generated only if that receipt belongs to the student on this access token. Another student's receipt ID will be rejected.

A2.4 OTP (staging only)

The API can issue a local OTP challenge. Delivery is not real SMS. The test code is **246810**. After five wrong attempts the challenge locks. Do not use this as a production login.

Part B — Admin Guide

Staff work in Frappe Desk after `/login`. Use Awesome Bar (search at the top) to open any DocType by name, for example **Education Institution Node** or **Student Fee Demand**.

Do not give routine school staff the System Manager role. Use the baseline roles below.

B1. Desk basics that apply everywhere

- Save stores a draft. Submit locks the business action (confirm, convert, post payment).
- Many masters use status: **Draft** **Published** / **Approved** / **Active** **Locked** / **Retired** / **Cancelled**.
- Version (track changes) is on for critical records. Do not hard-delete history.
- Never change a submitted financial document in place. Reverse with a credit note / refund.
- Never set a payment to Paid because a browser redirected to "success." Confirm the provider event and the Payment Entry.
- Guest portal APIs are public but scoped by resume token or access token. Do not paste tokens into tickets or chat.

B1.1 Baseline roles

Role	May do	Must not do
Applicant / Guardian	Own portal draft, documents, dues, receipts	Any staff record or export
Admissions Officer	Enquiries, applications, documents, eligibility	Merit publish, refunds, concessions, role changes
Academic Officer	Academic masters, offerings, intake, sections	Financial posting, unmasked identity export
Finance Officer	Demands, invoices, payments, reconciliation	Unapproved refunds, write-offs, concessions
Registrar	Student identity, confirmation, conversion, corrections	Credential admin, unrestricted exports
Institution Administrator	Institution-scoped setup and users	Bypass audit or site isolation
System Manager	Technical configuration	Routine business approvals

Site isolation is absolute: one institution = one Frappe site = one database.

B1.2 Suggested Desk search list (pin these)

Institution: Education Institution Node, Institution Structure Version **Academic:** Academic Year, Academic Term, Academic Session Policy, Academic Calendar, Program, Program Version, Program Offering, Class Offering, Academic Section, Curriculum Version, Subject Offering, Program Intake, Faculty Assignment, Timetable Slot, Timetable Entry **CRM / admissions:** CRM Lead, Admission Application Form Version, Admission Application Draft, CRM Application Handoff, Student Applicant, Eligibility Rule Set, Eligibility Evaluation, Merit Configuration, Merit Run, Admission Seat Matrix, Seat Allocation Round, Seat Offer, Admission Confirmation, Admission Student Conversion **Identity:** Student, Student Identity Profile, Student Identity Issuance, Guardian, Student Guardian Relationship, Communication Consent, Student Correction Request, Duplicate Candidate **Documents:** Student Document Type, Document Requirement Matrix, Student Document, Document Verification, Admission Application Document **Fees:** Education Fee Code, Education Fee Policy Version, Student Fee Demand, Student Fee Payment, Student Fee Refund, Fee General Ledger Reconciliation, Sales Invoice, Payment Entry **Portal:** Student Portal Access, Student Portal Notice

B2. Institution setup (do this first)

Owner: Institution Administrator Records: Education Institution Node, Institution Structure Version

B2.1 Build the tree

Create nodes from the top down. The code is the document name (uppercased).

Node type	Parent required?	Typical example
University	No (root only)	The school trust / university
Campus	University or higher group	Main campus
College	Campus (or higher group)	High school / college
Department	College	Science / Primary

Rules:

- Only a University may have no parent.
- Parent must be a group node (**is_group**).
- Child type must sit below the parent (University Campus College Department).
- **inactive_from** cannot be before **active_from**.
- Status: **Active**, **Inactive**, **Locked**.
- Link **company** (ERPNext Company) and currency on the node that owns accounting.

Do not delete a node that transactions already reference. Deactivate or lock it.

B2.2 Publish a structure version

Create Institution Structure Version against the university node. Status: **Draft** **Published**. This is the snapshot admissions and reporting should treat as the live hierarchy.

B3. Academic year, calendar, program, class, intake

Owner: Academic Officer

Do this in order. Later admissions and fees depend on these names.

B3.1 Year, term, policy, calendar

1. Academic Year (Education) — e.g. **2026-27**.
2. Academic Term under that year.
3. Academic Session Policy — **Draft** **Published** **Locked** when admissions start.
4. Academic Calendar linked to the year and institution node. Status: **Draft** / **Published** / **Locked** / **Cancelled**.

5. Academic Calendar Day rows: **Working**, **Holiday**, **Exam**, **Event**.

B3.2 Program chain

Keep these separate. Do not clone a new Program every year.

Program	reusable identity (e.g. Class 6 / B.A.)
→ Program Version	published academic structure for a year
→ Curriculum Version + Curriculum Course rows	(Core / Elective / Ability Enhancement / Skill Enhancement / Value Added)
→ Program Offering	that version offered on a campus for a session
→ Class Offering	the class instance
→ Academic Section	section (e.g. 6-A) + optional class teacher
→ Subject Offering	course taught in that class

Statuses you will use:

Record	Statuses
Program Version	Draft / Published / Cancelled
Program Offering	Draft / Open / Locked / Closed
Class Offering	(configured per record; lock after timetable is live)
Academic Section	Active / Inactive / Locked
Subject Offering	Draft / Open / Locked / Closed

A subject offering must be Open or Locked before you can timetable it.

B3.3 Intake and reservation

1. Create Student Category values you will use (e.g. General).
2. Create Program Intake on the Program Offering.
3. Add Category Intake child rows. Category capacities must sum to Total Capacity.
4. Submit. Status becomes **Approved**. Cancel sets **Cancelled**. Later lock when the cycle must not change.

B3.4 Faculty and timetable

1. Create an Instructor.
2. Create Faculty Assignment (Active) on a Subject Offering.
3. Create Timetable Slot (weekday + time).
4. Create Timetable Entry (code is the name) linking subject, section, slot, instructor, optional room.

Clash detection rejects:

- same section in the same slot
- same faculty in the same slot
- same room in the same slot

Cancelled entries are ignored. Use this in the demo to show the system blocking a double-book.

B4. Open admissions (forms, CRM, handoff)

Owners: Institution Administrator (form), Admissions Officer / counsellor (enquiry)

B4.1 Publish the application form

Create Admission Application Form Version:

- Program and Academic Year
- **form_schema** JSON (fields the portal/API may expose)
- Status Published

The guardian portal lists only Published forms. Retire old versions; do not edit a live published form in place if families already started drafts.

B4.2 How an enquiry appears

Two equally valid paths:

Path 1 — Guardian starts on the portal `save_application_draft` creates CRM Lead + Admission Application Draft (**Draft**). Counsellor sees the lead in Frappe CRM / Desk.

Path 2 — Counsellor captures a walk-in Create CRM Lead in CRM (name, mobile, email, status). Then create the draft and handoff yourself.

CRM is for pre-application relationship only. It is not the permanent student master.

B4.3 Review the draft

Open Admission Application Draft:

Field	Meaning
Form Version	Published form used
CRM Lead	Enquiry
Status	Draft / Submitted / Abandoned
Draft Payload	JSON of guardian answers
Student Applicant	Filled after handoff
Resume Token Hash	Cannot be used to impersonate the family

Also open Admission Application Document (scan status) and Admission Payment Attempt (**Pending** / **Paid**).

B4.4 Handoff to Student Applicant

Create CRM Application Handoff:

- CRM Lead
- Program, Academic Year, Academic Term
- Form Version and Application Draft
- Handoff date

Submit. The system:

1. Creates or reuses Student Applicant (Education)
2. Marks the CRM Lead converted
3. Sets the draft to Submitted and stamps `submitted_on`
4. Sets handoff status to Application Created

Rules:

- One active handoff per CRM Lead
- Repeat submit is idempotent on the applicant email
- After submit, the guardian can no longer edit that draft

B5. Documents and student identity

Owners: Document Verifier / Admissions Officer / Registrar

B5.1 Configure what is required

1. Student Document Type (e.g. Birth certificate) — Active.
2. Document Requirement Matrix — program / category / Active.
3. Document Rejection Reason catalogue.

B5.2 Student document lifecycle

Uploaded → Scanning → Scan Passed / Scan Failed
→ Pending Verification → Verified / Rejected / Expired / Replaced

Staff records:

DocType	When to use
Student Document	The file metadata for an applicant or student
Document Scan Result	Passed / Failed scan evidence
Document Verification	Verified or Rejected + reason
Document Replacement Request	Draft Approved / Rejected
Document Expiry Review	Open Expired / Resolved

Portal uploads during application land on Admission Application Document, not yet the permanent Student Document. After conversion, registrar/staff attach or migrate verified files onto the student.

Private files only. Do not email full documents.

B5.3 Identity after the child exists as an applicant or student

DocType	Purpose
Student Identity Profile	Canonical profile (Draft / Active / Locked / Merged / Cancelled)
Student Identity Issuance	Immutable student number + enrollment number (submit to Issue)
Student Guardian Relationship	Link to Education Guardian
Communication Consent	Channel consent history
Student Category History	Category changes with dates
Student Status Change	Lifecycle history
Student Correction Request	Draft Approved / Rejected / Cancelled
Duplicate Candidate	Open Confirmed Duplicate / Dismissed — never silent merge
Student Data Access Log	View / Export / Mask / Correction
Student Privacy Export Request	Draft Approved / Rejected; default masked export

Correction of issued IDs requires an approved correction request. Full Aadhaar must not be a default identifier; if collected under a lawful basis, mask it and keep it off lists, logs, and URLs.

B6. Eligibility, merit, and seats

Owners: Admissions Officer (eligibility), Merit Operator / Approver (merit and seats)

B6.1 Eligibility

1. Publish Eligibility Rule Set for the program/year. `rules_json` must include `minimum_score`.
2. Create Eligibility Evaluation for the Student Applicant:
 - `score`
 - `result` `Eligible` or `Ineligible`
 - `explanation_json` that includes `minimum_score` so the decision is explainable

The server recomputes the expected result from the published rules. A mismatched result is rejected. Only Published rule sets can be used.

B6.2 Merit (immutable)

1. Merit Configuration — Active, with scoring / tie-breaker JSON.
2. Merit Run against that configuration, program, and year.
3. Add Merit Entry rows (applicant, category, scores, rank).
4. Submit the run. Status becomes Published. `published_on` is required.

You cannot cancel a published merit run. To correct it, create a new run and mark the old one Superseded. Regeneration is a new approved version, never an overwrite.

B6.3 Seat matrix and offers

1. Admission Seat Matrix — Program Offering + Student Category + capacity. Status `Draft` / `Locked` / `Retired`.
2. Seat Allocation Round on a Published Merit Run. Publish the round (`Draft` / `Published` / `Closed`).
3. Seat Offer per applicant in that round:
 - `Offered` — issued
 - `Accepted` — requires `accepted_on`; capacity is locked with a row lock
 - `Waitlisted`
 - `Expired` / `Cancelled`

Rules:

- Offers require a Published allocation round
- One offer per applicant per round
- Submit of an Accepted offer fails if accepted count already equals capacity

This is the slide where you show “the system will not sell the 21st seat.”

B7. Confirm admission and create the student

Owner: Registrar / Admission Approver

B7.1 Admission Confirmation

Create Admission Confirmation on a submitted Accepted Seat Offer.

Before submit you must have:

- `document_gate_passed`
- `fee_gate_passed`
- `confirmed_on`

Status becomes `Confirmed`. Waitlisted offers are rejected. Missing gates are rejected.

B7.2 Admission Student Conversion

Create Admission Student Conversion on that submitted confirmation. Set `conversion_date`. Submit.

The system will:

1. Create or reuse Student from the applicant
2. Create or reuse submitted Program Enrollment
3. Create or link Student Identity Profile
4. Issue Student Identity Issuance (student number + enrollment number)
5. Set Student Applicant `application_status` to `Admitted`
6. Set conversion status to `Converted`

Repeating conversion returns the same student and enrollment. A second active conversion for the same applicant is rejected.

After this, issue Student Portal Access (hash the raw token, store only the hash, set `expires_on`) and send the family `/student-portal?access=<raw-token>`.

B8. Fees, collection, refunds, reconciliation

Owner: Finance Officer / Cashier / Finance Approver Accounting truth: ERPNext Sales Invoice, Payment Entry, credit note, General Ledger

B8.1 Configure fee policy

1. Fee Category (Education) if needed.
2. Education Fee Code — Draft / Active / Retired.
3. Education Fee Policy Version — program, year, fee code, amounts. Publish it (Draft / Published / Retired).
4. Education Fee Installment child/related rows if the policy is split.
5. Student Fee Adjustment for Concession / Scholarship / Fine / Waiver (Draft / Approved / Rejected). Gross fee stays visible; net is derived.

Net formula enforced on the demand:

```
net = gross - concession - scholarship + fine - waiver
```

Net cannot be negative. Generated demands must have a Sales Invoice whose grand total equals net.

B8.2 Generate a demand

1. Ensure the student has a submitted Program Enrollment.
2. Create / generate Student Fee Demand:
 - student, enrollment, published policy
 - amounts and due date
 - linked submitted Sales Invoice
3. Submit. Status becomes Generated.

Do not mark a demand Generated without an invoice.

B8.3 Collect payment

Online (portal or Desk):

1. Create a payment attempt (Admission Payment Attempt for application fee, Student Portal Payment Attempt or finance collection for tuition).
2. Verify the provider order / capture (fake Razorpay in staging).
3. Create and submit ERPNext Payment Entry against the Sales Invoice.
4. Create and submit Student Fee Payment (Online) with the same amount and Payment Entry.
5. Receipt number is issued (FEE-REC-... if not supplied).

Offline (cashier):

1. Collect cash/cheque/UPI at the counter.
2. Post Payment Entry.
3. Submit Student Fee Payment with `collection_type = Offline` and `approved_on` filled.

Duplicate `provider + provider_payment_id` is rejected. Browser “success” is not enough.

B8.4 Refund

1. Create Student Fee Refund from the posted payment. Amount > 0.
2. Approve (`approved_on` required). Status Approved.
3. Post a return Sales Invoice (credit note) and a refund Payment Entry for the same amount.
4. Submit the refund. Status becomes Posted.

Duplicate provider refund IDs are rejected. Unapproved refunds cannot be posted.

B8.5 Settlement and GL

1. Payment Settlement Import — Imported / Reconciled / Mismatch.
2. Fee General Ledger Reconciliation against the demand and Sales Invoice — Draft / Reconciled / Mismatch.

A mismatch must stay visible. Do not force Reconciled when totals disagree.

B8.6 Application fee vs tuition fee

Fee	Who pays	Record	Amount in current portal
Application fee	Guardian during apply	Admission Payment Attempt	₹500 fixed
Tuition / school fee	Student portal or cashier	Student Fee Demand + Student Fee Payment	Policy net amount

Application-fee attempts do not by themselves create the tuition Sales Invoice. Tuition starts after the student exists and finance generates a demand.

B9. Notices, portal access, and helping a family

B9.1 Notices

Student Portal Notice

- Title, message, published_on
- Audience: All Students / Guardians / Staff
- Status: Draft / Published / Archived
- Optional expires_on

Only Published notices for All Students or Guardians appear on `/student-portal`.

B9.2 Issue or revoke portal access

1. Generate a high-entropy token (do not type a guessable ID).
2. Store SHA-256 hash in Student Portal Access. `status = Active`, set `expires_on`.
3. Send the family the raw token once, as a link.
4. To cut off access: set status `Revoked` or let it `Expired`.

`last_used_on` updates when the portal loads.

B9.3 Assisted entry

If a guardian cannot complete the phone form, a counsellor may capture the CRM Lead and draft in Desk. Treat this as assisted entry: keep the guardian's consent, do not invent documents, and do not mark fees paid without money.

B10. Security, privacy, audit, migration

Owner: System Manager / Auditor / Institution Administrator

B10.1 Daily security habits

- Named accounts, not shared `Administrator`, for school staff
- MFA for finance, merit approvers, auditors, and platform operators (production policy)
- Private files only for applicant/student documents
- Masked exports by default; unmasked export needs an approved Student Privacy Export Request
- Webhooks must verify signatures; replayed events are rejected
- Do not log passwords, OTPs, payment signatures, or full Aadhaar

B10.2 What to show an auditor

- Frappe Version on tracked masters and identity
- Student Data Access Log
- Payment Provider Event (**Received** / **Processed** / **Duplicate** / **Rejected**)
- Submitted merit runs that cannot be cancelled
- Fee GL reconciliation Reconciled vs Mismatch

B10.3 Trial migration (synthetic only)

Templates in `docs/operations/migration-templates/`:

- `students.csv`
- `guardians.csv`
- `fee_opening_balances.csv`

Validate with `scripts/migration/validate-trial-load.py`. Do not import identifiable student data without explicit authorization. Opening balances need finance sign-off before posting.

B10.4 Health checks

```
GET /api/method/university_erp.api.health.live
GET /api/method/university_erp.api.health.ready
```

Both should return HTTP 200 before a demo.

B11. Recommended operating calendar (one admission cycle)

When	Who	Action
T-6 weeks	Institution + Academic	Tree, year, program, class, sections, timetable
T-5 weeks	Academic	Intake approved and locked
T-4 weeks	Finance	Fee codes and published policy
T-3 weeks	Admissions	Publish form, eligibility rules, document matrix
T-2 weeks	Admissions	Counsellor training; portal smoke test
Cycle open	Guardians	Apply, upload, pay ₹500
Daily	Admissions	Handoff, scrutiny, document verification
After close	Merit	Publish merit run; allocation rounds; offers
Offer window	Registrar + Finance	Gates, confirmation, conversion, portal links
After join	Finance	Generate demands; collect; reconcile
Monthly	Finance + Auditor	Settlement import, GL check, access-log sample

Part C — Demo Presentation Flow

Use this as a 25–35 minute stakeholder demo. One presenter drives the phone (or a narrow browser window). A second window stays on Desk. Use synthetic data only.

C1. Goal of the demo

Leave the room believing three things:

1. A parent with a basic phone can start and finish an application without training.
2. Staff can take that enquiry all the way to a real student record without double-creating anyone.
3. Money is accounted for in ERPNext, including “do not pay twice.”

Do not claim live Razorpay, SMS, or production go-live. Say clearly: “This staging build uses safe fake payment and scan adapters so we can show the controls before real money moves.”

C2. Prep (30 minutes before the room)

1. Confirm `/api/method/university_erp.api.health.ready` is 200.
2. Confirm a Published Admission Application Form Version exists.
3. Confirm Academic Year, Program, Program Offering, Class, Student Category, and Intake exist.
4. Confirm a Published Eligibility Rule Set, Merit Configuration, Seat Matrix (small capacity, e.g. 1 or 2), and Fee Policy exist — or be ready to show existing proof records.
5. Log into Desk as a named staff user (not root if you can avoid it).
6. Open a second browser profile or incognito for the guardian so cookies do not mix.
7. Narrow the guardian window to a phone width (~390px).
8. Clear the guardian profile's `localStorage` if you need a clean “first time” story (`university_erp_guardian_application`).
9. Have a sample PDF and a JPG under 5 MB.
10. Write the student-portal URL pattern on a speaker note: `/student-portal?access=...`

If the site has existing proof data ([P3.1 Proof University](#), `EDU-STU-2026-00002`, etc.), you may fast-forward staff setup and spend time on the live guardian + conversion + fee story.

C3. Timed script

Act 0 — Frame (2 minutes)

Say: “This is an Education ERP for a township high school that must also work for a university later. Families use a Hindi/English phone portal. Staff use Desk. Accounts use ERPNext so fees are not a spreadsheet.”

Show: one slide or the opening of this guide's flow diagram. Then switch to the live site.

Do not tour every DocType.

Act 1 — Guardian applies on a phone (8 minutes)

URL: `/guardian-admission`

Minute	You do	You say
0:00	Toggle English हिन्दी English	“Same flow in both languages. Large targets, few fields.”
0:30	Register: mobile, guardian, child	“We start with mobile and names only. No password yet.”
1:00	Point at “Saved / Saved to school”	“The form autosaves on the phone and to the school.”
1:15	Class 6, previous school, Next	“Class list is the school's published offering, not a free-text mess.”
2:00	DOB, address, consent	“Consent is a required gate, not a footer.”
2:45	Upload birth certificate + photo	“Private upload. File type and size checked. Scan must pass.”
4:00	Pay ₹500, then tap again	“Second tap does not create a second order. That is the duplicate-pay protection.”

Minute	You do	You say
5:00	Check payment status	"Families are told: if money is cut, check status, do not pay again."
6:00	Status step	"The family is done. Rank, offer, and admission are school work."
6:30	Optional: toggle airplane mode, type, go back online	"Offline draft survives a dropped village network."

If payment stays Pending: that is acceptable. Explain that capture is server-side and show the Admission Payment Attempt in Desk.

Act 2 — Staff see the enquiry (4 minutes)

In Desk:

1. Awesome Bar CRM Lead open the child just created.
2. Open Admission Application Draft — status Draft, payload, linked lead.
3. Open Admission Application Document — Scan Passed.
4. Open Admission Payment Attempt — order id, status.

Say: "Nothing the parent typed is trapped in a WhatsApp chat. The counsellor already has an enquiry, a draft, documents, and a fee attempt."

Then submit CRM Application Handoff.

Show: Student Applicant created; draft now Submitted; lead converted.

Say: "Handoff is idempotent. If the counsellor clicks twice, we do not create two applicants."

Act 3 — Eligibility, merit, seat (5 minutes)

Keep this tight. Prefer pre-seeded rules if the clock is short.

1. Open Eligibility Evaluation (or create one). Show Eligible + explanation JSON / minimum score.
2. Open a Published Merit Run and one Merit Entry (rank).
3. Attempt to cancel the merit run if you want the error: published merit runs are immutable.
4. Open Admission Seat Matrix (capacity 1 or 2).
5. Open Seat Offer Accepted. Mention waitlist as the other path.
6. Optional punch: try a second Accepted offer on a full matrix and show the capacity error.

Say: "Merit is a published snapshot. We never silently rewrite yesterday's list. Seats cannot be oversold even if two clerks submit at once."

Act 4 — Confirmation and one student (4 minutes)

1. Open Admission Confirmation. Tick document gate + fee gate, set confirmed_on, Submit.
2. Open Admission Student Conversion, Submit.
3. Show Student, Program Enrollment (submitted), Student Identity Issuance.
4. Submit conversion again (or explain the proof): same student, same enrollment.

Say: "This is the permanent identity. Applicant becomes one student and one enrollment. That is what the rest of the school will use for fees and the student portal."

Act 5 — Fees that an accountant will trust (6 minutes)

1. Open Education Fee Policy Version (Published) and the amounts.
2. Open Student Fee Demand + linked Sales Invoice (submitted, totals match).

3. Show Student Fee Payment + Payment Entry (online and, if present, offline).
4. Show a receipt number.
5. Open Student Fee Refund path or an existing posted refund (credit note + refund Payment Entry).
6. Open Fee General Ledger Reconciliation — Reconciled. Mention Mismatch is a first-class status.

Say: “Education rules live in our app. The money lives in ERPNext. If settlement and the ledger disagree, the system keeps a mismatch instead of hiding it.”

Act 6 — Student portal (4 minutes)

1. Issue or reuse Student Portal Access.
2. Open `/student-portal?access=...` on the phone window.
3. Show student name, dues, receipts, notices, documents.
4. Download a receipt PDF.
5. Tap Pay on a due (or show already Paid).
6. Toggle Hindi.

Say: “After admission the family does not need Desk. They get a time-limited link: dues, pay, receipt, notices. Access can be revoked.”

Act 7 — Close (2 minutes)

Show this table on a slide or verbally:

Question	Answer in this build
Can a low-literacy parent apply?	Yes — six steps, EN/HI, autosave
Can seats be oversold?	No — locked capacity
Can we rewrite merit?	No — new version only
Can we create two students?	No — idempotent conversion
Can we double-charge?	No — idempotency keys + unique provider payment
Is money in a real ledger?	Yes — Sales Invoice / Payment Entry / GL
Is this live production?	Not yet — staging, fake providers, UAT pending

Next ask: authorize human UAT, then real sandbox credentials, then P9.1 pilot.

C4. Demo variants

C4.1 10-minute executive cut

1. Guardian register documents pay (3 min)
2. Handoff Student Applicant (1 min)
3. Existing merit + accepted offer (1 min)
4. Existing Student + invoice + receipt (2 min)
5. Student portal receipt download (2 min)
6. Close lines (1 min)

C4.2 45-minute operational workshop

Add after Act 3:

- Create a colliding Timetable Entry and show the clash error
- Open Duplicate Candidate and state that merge is never silent
- Open Student Correction Request

- Walk Student Privacy Export Request (masked vs privileged)
- Show **live** / **ready** health endpoints
- Walk the UAT script IDs **UAT-001-UAT-009** as a checklist, not a live execution

C4.3 Things that will stall a room — avoid them

- Logging in as Administrator and wandering Awesome Bar
- Editing unpublished form JSON live
- Real card data or production DNS
- Claiming attendance, exams, or LMS
- Showing Aadhaar or unmasked exports
- Starting a second guardian application “to be safe”

C5. Talking points by audience

Audience	Emphasize	Skip
Principal / trustee	Guardian simplicity, one student identity, Hindi	DocType names
Admission clerk	CRM lead, draft, documents, handoff	GL theory
Accountant	Invoice = demand, Payment Entry, refund, mismatch	Portal CSS
IT	Site-per-institution, no core forks, fake-to-real adapters	Merit formula detail
Parent representative	Six steps, pay once, receipt PDF	Desk

Part D — Feature catalogue

Everything below exists as a custom DocType or portal/API behavior in **university_erp**, unless marked as an upstream Education / CRM / ERPNext record that this product reuses.

D1. Institution and academic

Feature	Record	What it does
Institution tree	Education Institution Node	University / Campus / College / Department nested set
Structure snapshot	Institution Structure Version	Draft / Published / Cancelled
Session rules	Academic Session Policy	Draft / Published / Locked / Cancelled
Calendar	Academic Calendar + Academic Calendar Day	Working / Holiday / Exam / Event
Program identity	Program (Education)	Reusable program
Versioned curriculum	Program Version, Curriculum Version, Curriculum Course	Published structure + course classification
Offering	Program Offering, Class Offering, Academic Section	Session-bound class/section
Subject	Subject Offering	Draft / Open / Locked / Closed
Intake	Program Intake, Category Intake	Category totals must equal capacity
Faculty	Faculty Assignment, Instructor (Education)	Active / Inactive

Feature	Record	What it does
Timetable	Timetable Slot, Timetable Entry	Clash detection on section, faculty, room

D2. CRM and application

Feature	Record / API	What it does
Enquiry	CRM Lead	Counsellor pipeline
Form version	Admission Application Form Version	Draft / Published / Retired
Guardian draft	Admission Application Draft + <code>save_application_draft</code>	Create/update, hashed resume token
Resume	<code>get_application_draft</code>	Load draft by token
Published forms	<code>get_application_context</code>	Portal bootstrap
Documents	Admission Application Document + <code>upload_application_document</code>	Private store + scan state
Application fee	Admission Payment Attempt + create/check/confirm APIs	₹500, idempotent
Handoff	CRM Application Handoff	Lead Student Applicant, submit draft
Applicant	Student Applicant (Education)	Official application record

D3. Eligibility, merit, seats, conversion

Feature	Record	What it does
Rules	Eligibility Rule Set	Published JSON, minimum score
Decision	Eligibility Evaluation	Eligible / Ineligible + explanation
Scoring config	Merit Configuration	Draft / Active / Retired
Merit list	Merit Run, Merit Entry	Publish; cannot cancel
Capacity	Admission Seat Matrix	Draft / Locked / Retired
Round	Seat Allocation Round	Draft / Published / Closed
Offer	Seat Offer	Offered / Accepted / Waitlisted / Expired / Cancelled
Confirm	Admission Confirmation	Document + fee gates
Convert	Admission Student Conversion	One Student + one Program Enrollment + identity issuance

D4. Identity and documents

Feature	Record	What it does
Profile	Student Identity Profile	Active identity
Numbers	Student Identity Issuance	Immutable student / enrollment numbers
Guardian link	Student Guardian Relationship, Guardian	Family link
Consent	Communication Consent	Channel consent
Histories	Student Category History, Student Status Change	Dated changes

Feature	Record	What it does
Corrections	Student Correction Request	Approval workflow
Dedupe	Duplicate Candidate	Review only
Document types	Student Document Type, Document Requirement Matrix	What is required
Student files	Student Document, Document Scan Result, Document Verification	Scan + verify
Replace / expiry	Document Replacement Request, Document Expiry Review	Lifecycle
Privacy	Student Privacy Export Request, Student Data Access Log	Approval + audit

D5. Fees and accounting

Feature	Record	What it does
Catalogue	Education Fee Code	Draft / Active / Retired
Policy	Education Fee Policy Version, Education Fee Installment	Published amounts / schedule
Adjustments	Student Fee Adjustment	Concession / Scholarship / Fine / Waiver
Demand	Student Fee Demand + Sales Invoice	Net must match invoice
Collection	Student Fee Payment + Payment Entry	Online / Offline, receipt
Portal pay	Student Portal Payment Attempt	Retry-safe tuition order
Events	Payment Provider Event	Order / capture / callback / refund
Refund	Student Fee Refund + credit note + refund PE	Approved then Posted
Settlement	Payment Settlement Import	Imported / Reconciled / Mismatch
GL	Fee General Ledger Reconciliation	Reconciled / Mismatch

D6. Portals and notices

Feature	Where	What it does
Guardian PWA	/guardian-admission	Six-step EN/HI apply
Student PWA	/student-portal	Dues, pay, receipts, documents, notices
Access	Student Portal Access	Hashed token, expiry, revoke
OTP contract	Student Portal OTP Challenge	Fake local OTP, lock after 5
Notices	Student Portal Notice	Published to families
Receipt PDF	download_student_receipt	Scoped to the token's student
Health	/api/method/university_erp.api.health.live / ready	Staging/ops probe

D7. Integrations (adapters)

Concern	Staging adapter	Production target (gated)
Payments	Fake Razorpay	Razorpay (institution-owned merchant)

Concern	Staging adapter	Production target (gated)
SMS	Fake MSG91	MSG91 after DLT / sender ownership
Email	Fake SMTP	Hostinger Business Email SMTP
Files	Fake R2	Cloudflare R2 private bucket
Antivirus	Fake ClamAV	Approved scanner

All adapters are behind ports so domain code does not hard-code a vendor.

D8. Portal API cheat sheet

All under `/api/method/university_erp.api.portal.*` (guest, token-scoped).

Method	Purpose
<code>get_application_context</code>	Published forms
<code>save_application_draft</code>	Create/update draft + CRM lead
<code>get_application_draft</code>	Resume
<code>upload_application_document</code>	Private upload + scan
<code>create_application_payment</code>	₹500 order
<code>check_application_payment</code>	Status only
<code>confirm_application_payment</code>	Capture once
<code>get_student_portal_snapshot</code>	Student view
<code>download_student_receipt</code>	PDF
<code>create_student_payment</code>	Tuition order
<code>confirm_student_payment</code>	Capture + Payment Entry + receipt
<code>check_student_payment</code>	Status
<code>request_student_otp / verify_student_otp</code>	Fake OTP contract

E. Quick reference — statuses

Record	Statuses
Education Institution Node	Active / Inactive / Locked
Institution Structure Version	Draft / Published / Cancelled
Academic Session Policy / Calendar	Draft / Published / Locked / Cancelled
Program Version	Draft / Published / Cancelled
Program Offering / Subject Offering	Draft / Open / Locked / Closed
Academic Section	Active / Inactive / Locked
Program Intake	Draft / Approved / Locked / Cancelled
Admission Application Form Version	Draft / Published / Retired
Admission Application Draft	Draft / Submitted / Abandoned
Admission Payment Attempt	Created / Pending / Paid / Failed / Cancelled
Admission Application Document	Uploaded / Scanning / Scan Passed / Scan Failed

Record	Statuses
CRM Application Handoff	Pending / Application Created / Failed
Eligibility Rule Set	Draft / Published / Retired
Eligibility Evaluation	Eligible / Ineligible
Merit Configuration	Draft / Active / Retired
Merit Run	Draft / Published / Superseded
Admission Seat Matrix	Draft / Locked / Retired
Seat Allocation Round	Draft / Published / Closed
Seat Offer	Offered / Accepted / Waitlisted / Expired / Cancelled
Admission Confirmation	Draft / Confirmed / Cancelled
Admission Student Conversion	Draft / Converted / Failed
Student Identity Profile	Draft / Active / Locked / Merged / Cancelled
Student Document verification	Pending Verification / Verified / Rejected / Expired / Replaced
Duplicate Candidate	Open / Confirmed Duplicate / Dismissed
Student Correction Request	Draft / Approved / Rejected / Cancelled
Education Fee Policy Version	Draft / Published / Retired
Student Fee Demand	Draft / Generated / Cancelled
Student Fee Payment	(submit sets Posted / receipt issued)
Student Fee Adjustment	Draft / Approved / Rejected
Student Fee Refund	Draft / Approved / Posted / Rejected
Payment Settlement Import	Imported / Reconciled / Mismatch
Fee General Ledger Reconciliation	Draft / Reconciled / Mismatch
Student Portal Access	Active / Revoked / Expired
Student Portal Notice	Draft / Published / Archived
Student Portal OTP Challenge	Pending / Verified / Expired / Locked

F. Honest limits for trainers

- Desk permissions on custom DocTypes are still primarily System Manager and Academics User in metadata. The role matrix is the target; full Role Permission Manager coverage is not signed off.
- There is no dedicated custom Workspace yet. Staff use Awesome Bar / module search.
- Guardian portal class list is hard-coded to Class 6–9 in the Vue app; live program offerings are not yet a dynamic dropdown.
- Application payment confirmation in the UI currently checks status; full browser-to-capture wiring is the fake-provider path, not Razorpay Checkout.js.
- Student portal Pay starts an order and polls status; a hosted checkout page is not implemented.
- Notifications are designed as an outbox; live SMS/email sending is not approved.
- Human UAT ([docs/quality/pilot-uat-script.md](#)) is still pending sign-off.

Train staff on the flows in this file, then execute UAT-001 through UAT-009 on synthetic data before any parent uses a real phone number.

G. Related documents

Document	Use
Pilot scope	What the first school is allowed to go live with
UX and localization	Language and low-literacy rules
Role matrix	Permission boundaries
Pilot UAT script	Formal test cases
Marketing features	External feature language
Current implementation status	What is actually verified
AGENTS.md	Engineering contract

Guide version: 1.0 · Prepared 2026-08-17 · Matches implemented `university_erp` portals and DocTypes through Phase 8 staging.